Trenchscan: phone video to 3D, tested on a radiator

Jakob Wirén · October 2026 · Hand-in: repository, notebook `notebooks/01_sparse_reconstruction.ipynb`, this report.
References such as §2.4 point to notebook sections.

1 Goal

sizeless documents underground utilities from a phone video. A crew films an open trench with an iPhone Pro for about ten minutes; sizeless builds a metric 3D model, recognises pipes, fittings and couplings, and delivers CAD and GIS as-built plans, according to its website and Y Combinator page. To understand the work behind such a product, I rebuilt the first stage, from video to 3D, with standard tools on a stand-in scene: what a phone video gives classical reconstruction, where it breaks, and so what sizeless has to solve.

2 Setup

The stand-in for a trench is a white column radiator in my apartment. Its vertical tubes play the pipes. The capture shares the constraints of a trench: a modest phone (Oppo A38, 720p maximum), no markers on the surfaces, limited stand-off (furniture and a corner), and smooth, uniform surfaces. The right end stands in a corner and cannot be filmed, so before any processing it is clear the model will be weak on that side.



Figure 1. The three clips, 36–47 s each: top-down along the radiator with bank cards on the floor, a side view of the left end, and an arc from the left end round to the front. Only clip 3 is reconstructed.

Tape-measured ground truth: tube diameter 28.6 mm, tube spacing 55 mm centre to centre, radiator height 960 mm. The bank cards in clips 1 and 2 are a scale reference, not used yet (Section 7). Clip 3 goes through a blur filter and then standard structure-from-motion (COLMAP) on a laptop CPU, in about a minute; settings in §2.1–2.2.

3 Theory in brief

Follow one capture through the steps.

- **Filming creates parallax.** Walking along the trench, the camera sees the same pipes from a slightly different position in every frame. *Parallax* is the apparent shift of an object between two views taken from different places: near things shift more than far things, as a finger held up does when you close one eye and then the other. That shift is what reveals depth. It needs a *baseline*, a distance between the two camera positions; a camera that only turns on the spot gives none.
- **Features link the frames.** To use parallax, the software must find the same physical spot in different frames. It looks for *features*: small, distinctive patches such as a corner, a speck or a printed letter. The classic detector, SIFT, finds small spots that stand out from their surroundings and describes each by the pattern of brightness changes around it, so it can be recognised in the next frame. A smooth surface with soft shading has no such spots. On a clean white pipe there is nothing to hold on to.

- **Geometry finds the cameras and the points.** Some matches are wrong. All correct matches between two frames must agree with one single camera movement (*epipolar geometry*), so a voting method, RANSAC, keeps the matches that agree and drops the rest. *Structure-from-motion* (SfM, here COLMAP) then works out where the camera stood for each frame and where each matched spot sits in 3D, by intersecting the viewing rays from two or more camera positions (*triangulation*). It adds frames one at a time, placing each new camera from 3D points it already sees. Last, *bundle adjustment* fine-tunes all cameras and points together so that every 3D point, projected back into a frame, lands where it was seen. The distance left over, in pixels, is the *reprojection error*; well under a pixel means the model is consistent. sizeless runs the same ideas live, while filming, as SLAM (simultaneous localisation and mapping).
- **Scale is missing.** Images alone cannot tell a small pipe filmed close up from a large pipe filmed from further away: scaling the scene and the camera path together gives exactly the same frames (Hartley and Zisserman). The model has the right shape but no units, until something of known length is in view, such as a card, or a sensor measures the phone's movement in metres.
- **Measuring the video.** Two numbers describe each frame here. *Sharpness* is the variance of the Laplacian, a filter that responds to fine detail; blur smears detail away, so a blurred frame scores low. *Image speed* is how fast the picture moves across the screen, in pixels per second, taken as the median *optical flow* (Farnebäck), the estimated movement of each pixel between two frames.

4 Findings

4.1 The data (§1.1–1.4)

- **Coverage is one-sided.** Only the left side and the front are filmed, never the right end. The bank cards appear in clip 1 and early in clip 2, not in clip 3. Feet are in most of clip 1.
- **Sharpness does not follow speed.** In every clip, image speed and sharpness are barely related (§1.3). Clip 2's softest stretch is nearly still and shows bare wall: the sharpness score mixes up *blurry* with *featureless*.

4.2 The reconstruction of clip 3 (§2.1–2.4)

87 of the 107 frames kept by the blur filter register into one model, with 3,594 points at a reprojection error of 0.66 px: where frames register, the geometry is consistent (full statistics in §2.2).

- **The blur filter removes content, not only blur.** Many of the frames it rejects come from the slow opening seconds, where the view is mostly plain wall: sharp, but bare (§2.1).
- **Smooth surfaces are invisible to the model.** Regions without fine detail, the shaded tube faces and the wall, cover 57 % of an average frame but hold 4 % of the 3D points. The rest sit on edges, gaps, floor grain and the cards (Figure 2, top). The picture holds over a range of thresholds (§2.4).
- **The model breaks in the fastest, closest stretch.** All 20 unregistered frames lie in the fast close-up at the end; once the chain of frames breaks there, it does not recover. 3D points seen per frame drop from about 390 to about 90 as image speed rises (rank correlation −0.62, Figure 3). Speed alone does not separate the frames, though: every failed frame is slower than the fastest registered one, and speed rises together with closeness, so one clip cannot tell the two apart.
- **The geometry is right where there is texture.** The camera path is a smooth arc at a nearly constant distance. The points outline vertical edges and the top and bottom of the radiator, and leave the tube faces empty (Figure 4, top).

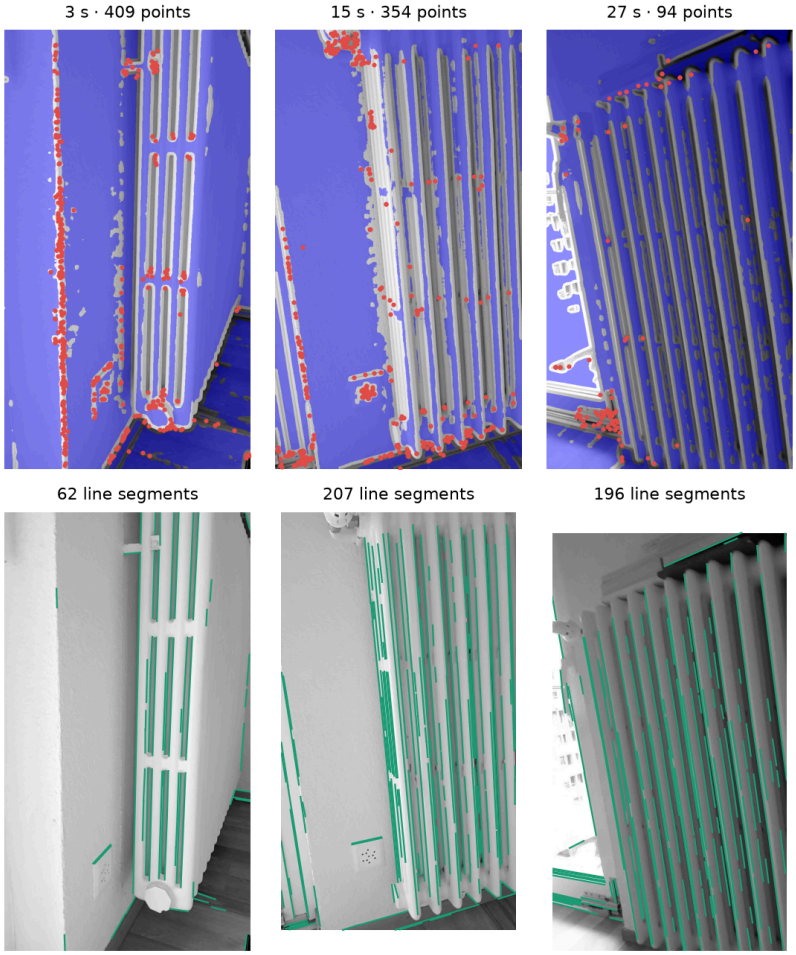


Figure 2. Three frames of clip 3. Top: regions without fine detail (blue) and keypoints that became 3D points (red). Bottom: line segments on the same frames, used in Section 5.

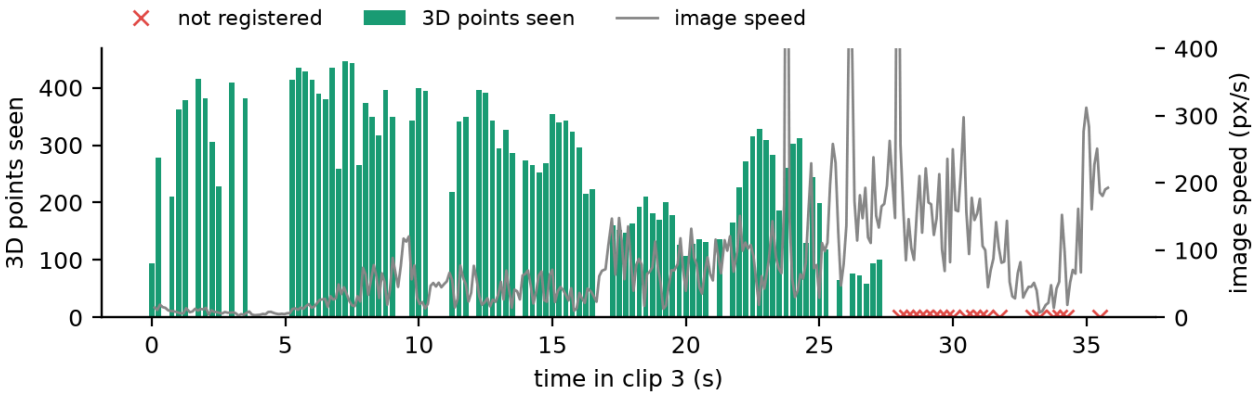


Figure 3. 3D points seen per kept frame of clip 3, frames that failed to register, and image speed (capped at 400 px/s).

5 Trial: lines instead of points (§3.1–3.4)

Why. The tube faces are smooth (Figure 2), but the gaps between the tubes are long straight edges. A feature has to stand out in every direction, but a spot on a straight edge looks like the next spot along it (the *aperture problem*), so SIFT finds points only where an edge ends or turns. A line as a whole, though, is easy to find. The question for the trial: are the tubes in the frames as lines?

Why this method.

- The SfM camera poses (§4.2) are reused as they are.
- A standard line-segment detector, LSD, finds the segments.

- The tubes are vertical. Seen from above, a vertical line is a single point. A segment in one frame and that frame's camera centre span a plane, and seen from above that plane is a ray from the camera through the line. The rays of one tube from many frames cross at one spot, so counting rays per grid cell (a Hough-style vote) finds the tubes without matching lines between frames.

General line mappers (Line3D++, LIMAP) handle lines in any direction but take more setup; the vertical shortcut is a few notebook cells, enough to answer the question.

Result.

- LSD finds the tube edges and gaps in every frame, including frames where few 3D points survived (Figure 2, bottom).
- From above, the votes form a row of compact, evenly spaced spots along the radiator front; the 3D points lie in the same area but scatter over it (§3.2).
- Along that row the votes repeat every 0.36–0.37 model units; the 3D points barely repeat (§3.3).
- The segment ends stop sharply at the top and bottom of the tubes, 6.0–6.2 units apart, so the segments span the full height. Spacing over height is a ratio, so it needs no scale: it gives 56–58 mm for the 960 mm radiator, against 55 mm on the tape.

The two approaches in one model. In Figure 4 each line runs from where its segments start to where they end. The 18 lines fill the radiator front the points left empty, and their ends meet the point bands at the top and bottom. There are more lines than tubes, though: 18 lines over 7.3 tube spacings, because one tube gives more than one line, its edges and the gaps beside it.

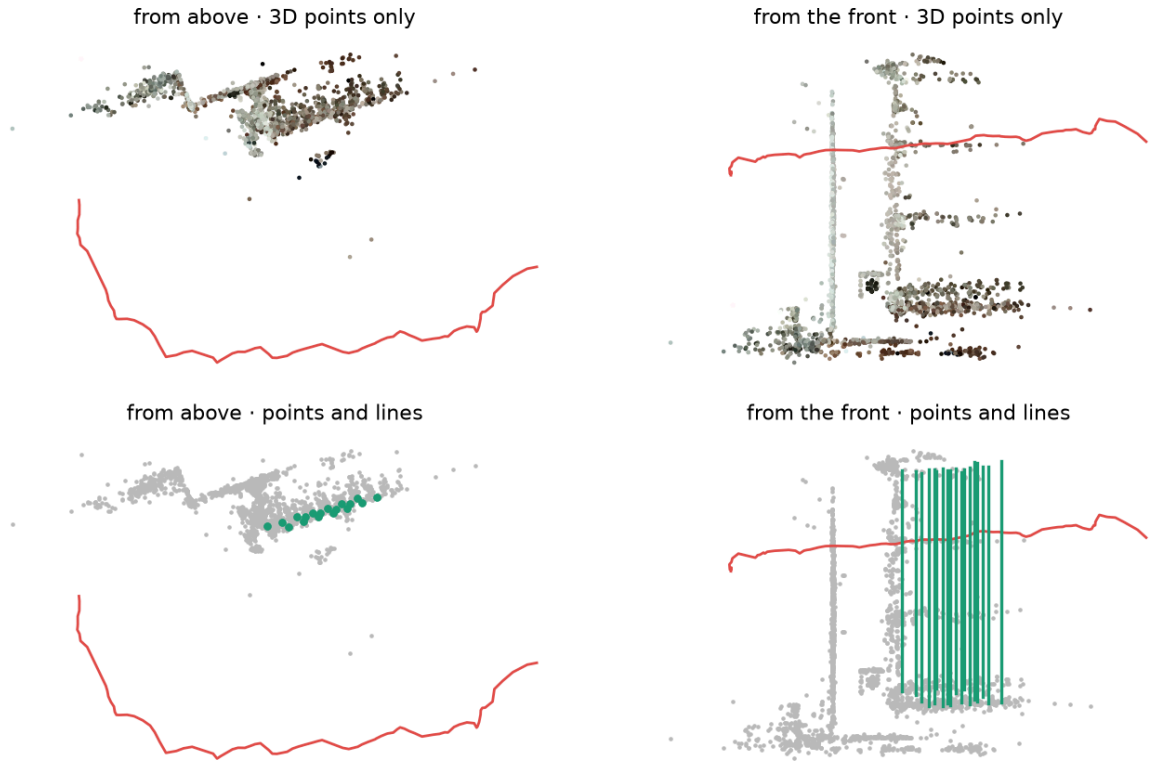


Figure 4. Top: the SfM model of §4.2, from above and from the front. Bottom: the same points (grey) with the vertical lines from the vote (green). Camera path in red; both rows use the vertical estimated from the lines (§3.2).

What it shows. The tube structure is in the video, as lines, and points miss most of it. The trial gives positions, not diameters: a tube's visible edge is where the line of sight grazes the cylinder, which moves with the camera, so a diameter needs a cylinder fitted to those edges. It relies on vertical tubes, while trench pipes mostly run horizontally along the trench, so the vote would have to run along the pipe direction, or a general line mapper takes over.

6 Discussion: what this means for a product like sizeless

1. **Poses are easy, the objects are hard.** The camera path is reasonable, but the pipes themselves, smooth and uniform, carry almost no points and therefore feature extraction is hard. In a trench, PVC, painted steel and water behave the same way. sizeless reports pipe diameters as an output; our evidence says surface points from classical features would not support that. Shape has to come from edges and silhouettes, as the lines in Section 5 do for position, or from methods that do not depend on texture.
2. **Guide the capture with image speed.** Speed probably correlates with the loss of points (§4.2); the sharpness score does not, and it discards sharp frames of bare surfaces. A capture app can measure optical flow live and warn before the frames are lost.
3. **Coverage is decided on site.** The unfilmed corner cannot be recovered later. Once a trench is backfilled, neither can the trench, so the app has to show what is missing while the crew is still there.
4. **Scale must be built in.** Our model has no units. sizeless captures with an iPhone Pro and builds a metric model in its SLAM engine; a card or something of known size in view is only a fallback, and only if it is in the clip that gets reconstructed, which it was not in this case.
5. **Real scenes move.** Feet already disturb clip 1. sizeless names wet, muddy trenches with moving excavators as what breaks off-the-shelf reconstruction.

Step	sizeless	This demo
Capture	Guided iPhone Pro app, about 10 min per trench, works without GPS	Unguided 36–47 s clips from a 720p phone
3D	Metric 3D model from a SLAM engine	Sparse SfM, about 3,600 points, no scale, static scene
Components	Pipe types, diameters, fittings, couplings	None
Measurements	Measurements and billable quantities	Tube spacing relative to height, from lines (Section 5); tape ground truth recorded
Output	CAD (DWG/DXF), BIM/GIS, crew voice notes anchored in 3D	Interactive 3D view

Sources for sizeless: website, YC page, App Store listing.

7 Future work

Each step follows from a finding and can be scored with data this project already has.

Finding	Next step	Evaluation
No scale (§2.2)	Scale from the bank card; combine clip 1, rotated, with clip 3	Tube spacing 55 mm and height 960 mm against the tape
Smooth tubes hold 4 % of points (§2.4)	Learned matching (SuperPoint + LightGlue) or learned reconstruction (MASt3R, VGGT)	Point share in smooth regions
Lines find the tubes, but only vertical ones and no diameter (§3.3)	Cylinders fitted to the tube edges; line mapping in any direction (LIMAP), tried on horizontal pipes	Diameter 28.6 mm; spacing 55 mm
Points fall as speed rises; the blur score drops sharp, bare frames (§2.1, §2.4)	Live speed warning during capture; select frames by speed and overlap instead of sharpness	Share of frames that register; points per frame
Moving objects in view (§1.2)	Mask people before matching	Reprojection error and outliers with and without masks

8 Limitations

Just capture more data, and learn by trial and error what types of data make reconstruction better. The smooth-region measure covers wall and tube faces together, because nothing segments the radiator yet. Its threshold was chosen by eye, so §2.4 checks a range of thresholds.